

REPUBLIC OF THE PHILIPPINES · DEPARTMENT OF EDUCATION

Grade 6 Mathematics

SECOND TERM EXAMINATION

SY 2026–2027 · MATATAG

50 points · Ratio & Proportion, Percent, Angles, Transformations, Volume
Includes Table of Specifications and Answer Key for the teacher

Second Term Examination

Name: _____ Section: _____ Date: _____ Score: _____/50

Read each item carefully. Show solutions for the problem-solving part.

I. Multiple Choice (1 pt each, 20 pts)

- Simplify the ratio 12 : 18.
A. 2:3 B. 3:2 C. 4:6 D. 6:9
- Find n: $\frac{2}{3} = \frac{n}{12}$
A. 6 B. 8 C. 9 D. 4
- 5 kg of rice costs ₱250. Cost of 8 kg?
A. ₱350 B. ₱400 C. ₱450 D. ₱500
- Divide 30 in the ratio 2:3. The parts are
A. 10 & 20 B. 12 & 18 C. 15 & 15 D. 6 & 9
- Write 35% as a fraction in lowest terms.
A. $\frac{35}{10}$ B. $\frac{7}{20}$ C. $\frac{1}{3}$ D. $\frac{3}{5}$
- Write 0.4 as a percent.
A. 0.4% B. 4% C. 40% D. 400%
- 25% of 80 =
A. 16 B. 20 C. 25 D. 40
- What percent of 60 is 15?
A. 15% B. 20% C. 25% D. 30%
- 9 is 30% of what number?
A. 27 B. 30 C. 90 D. 3
- A ₱500 item at 10% off costs
A. ₱450 B. ₱490 C. ₱400 D. ₱550
- Commission of 5% on ₱2,000 sales is
A. ₱50 B. ₱100 C. ₱200 D. ₱400
- Interest on ₱1,000 at 6% for 1 year is
A. ₱6 B. ₱60 C. ₱600 D. ₱16
- An angle of 95° is
A. acute B. right C. obtuse D. straight
- An angle that measures exactly 90° is
A. acute B. right C. obtuse D. reflex

15. A mirror image of a figure is a
 A. translation B. rotation C. reflection D. dilation
16. Sliding a figure without turning is a
 A. translation B. reflection C. rotation D. flip
17. The volume of a cube is found by
 A. $s \times 4$ B. s^2 C. s^3 D. $6s$
18. Volume of a $5 \times 4 \times 2$ cm box is
 A. 11 cm^3 B. 40 cm^3 C. 20 cm^3 D. 80 cm^3
19. 1 litre is equal to
 A. 10 mL B. 100 mL C. 1,000 mL D. 10,000 mL
20. A turn about a fixed point is a
 A. translation B. rotation C. reflection D. slide

II. Computation (2 pts each, 16 pts)

21. Find n : $\frac{5}{8} = \frac{n}{40} = \underline{\hspace{2cm}}$
22. Divide ₱1,200 in the ratio 1:2:3. = $\underline{\hspace{2cm}}$
23. 45% of 200 = $\underline{\hspace{2cm}}$
24. 24 is what percent of 96? = $\underline{\hspace{2cm}}$
25. Discount and sale price: ₱900 at 20% off. = $\underline{\hspace{2cm}}$
26. Simple interest: ₱6,000 at 4% for 2 years. = $\underline{\hspace{2cm}}$
27. Volume of a rectangular prism $12 \times 5 \times 3$ cm. = $\underline{\hspace{2cm}}$
28. Convert 4,500 mL to litres. = $\underline{\hspace{2cm}}$

III. Problem Solving (show solutions, 14 pts)

29. (3 pts) A recipe uses sugar and flour in the ratio 2:5. If 8 cups of sugar are used, how many cups of flour?

30. (3 pts) A ₱2,500 phone is on sale at 15% off. How much is the discount and the sale price?

31. (4 pts) Mang Jose deposited ₱10,000 at 3% simple interest per year. How much interest after 4 years, and what is his total amount?

32. (4 pts) A water container is 25 cm long, 20 cm wide, and 30 cm high. Find its volume in cm^3 and its capacity in litres.

Table of Specifications & Answer Key

Second Term Examination · 50 points

Table of Specifications

Competency (code)	Items	Pts	%
Ratio & proportion (6NA-IIa-1)	1, 2, 21, 29	8	16%
Direct/partitive proportion (6NA-IIb-2)	3, 4, 22	4	8%
Percent, rate, base (6NA-IIc-3)	5–9, 23, 24	9	18%
Percent in daily life (6NA-IId-4)	10, 11, 12, 25, 26, 30, 31	15	30%
Angles (6MG-IIe-5)	13, 14	2	4%
Transformations (6MG-IIf-6)	15, 16, 20	3	6%
Volume & capacity (6MG-IIg-7)	17, 18, 19, 27, 28, 32	9	18%
Total	32 items	50	100%

Answer Key – Part I (Multiple Choice)

1	2	3	4	5	6	7	8	9	10
A	B	B	B	B	C	B	C	B	A
11	12	13	14	15	16	17	18	19	20
B	B	C	B	C	A	C	B	C	B

Answer Key – Part II (Computation)

21) 25 22) ₱200, ₱400, ₱600 23) 90 24) 25% 25) discount ₱180, sale price ₱720 26) ₱480
(6000×0.04×2) 27) 180 cm³ 28) 4.5 L

Answer Key – Part III (Problem Solving)

29) $\frac{2}{5} = \frac{8}{f} \rightarrow 2f = 40 \rightarrow f = \mathbf{20 \text{ cups of flour.}}$

30) discount = $0.15 \times 2500 = \text{₱}375$; sale price = $2500 - 375 = \text{₱}2,125$.

31) $I = 10000 \times 0.03 \times 4 = \text{₱}1,200$; total = $10000 + 1200 = \text{₱}11,200$.

32) $V = 25 \times 20 \times 30 = \mathbf{15,000 \text{ cm}^3} = \mathbf{15 \text{ L}}$ (1 L = 1,000 cm³).